About the (squared) <u>correlation coefficient</u> and the (squared) <u>correlation quotient</u> (or equivalently <u>correlation ratio</u>)

I spoke quickly about these as more thorough discussion will follow up <u>after</u> this introductory lecture. Some students asked for more details after the lecture and I am including my discussion with them <u>here:</u>

If X and Y are two random variables and we want to use a transformation of X to approximate Y, a good way to measure the goodness of the approximation is $E(Y-g(X))^2$ where g denotes the transformation and we have assumed that $E(Y^2)<\infty$ and $E(g(X))^2<\infty$ holds.

We are interested in the <u>best</u> choice of $g(\,)$, i.e., in particular $g^*(X)=f(X)$ which is such that gives <u>minimal</u> value to $E(Y-g(X))^2$.

The claim was that $g^*(X)=f(X)=E(Y/X)$ "does the trick". The justification is utilizing the "iterative property" of the expected value and goes as follows: for <u>arbitrary</u> g such that $E(g(X))^2<\infty$ we have:

$$E(Y-g(X))^2=E(Y-f(X)+f(X)-g(X))^2=E(Y-f(X))^2+E(f(X)-g(X))^2$$
$$+\,2E[(Y-f(X))(f(X)-g(X))]$$

Now: $E[(Y-f(X))(f(X)-g(X))]=E_X[E(Y-f(X))(f(X)-g(X)/X]=$

$$=E_X[(f(X)-g(X))E((Y-f(X)/X)]$$

However $E(Y-f(X)/X)=E(Y/X)-f(X)=f(X)-f(X)=0$ hence the whole third summand on the RHS is 0

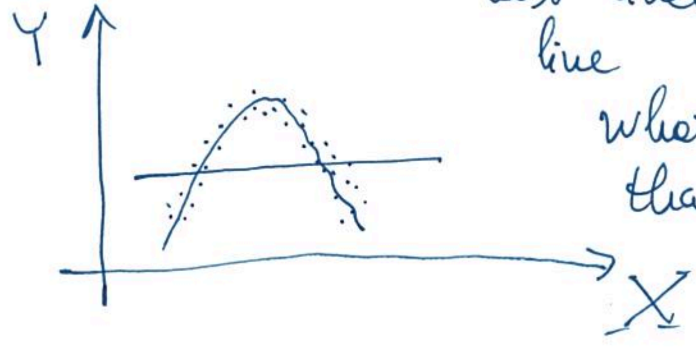
and we end up with

$$E(Y-g(X))^2 = E(Y-f(X))^2 + E(f(X)-g(X))^2 \geq E(Y-f(X))^2$$

Hence the "best" $g(X)$ is indeed $g^*(X) = f(X) = E(Y/X)$. This function $g^*(X)$ (the conditional expected value of Y given X) is nonlinear in general! For simplicity, sometimes we will be interested in the best approximation of Y with linear functional transformations of X, i.e., we want to find, in the class $g(X) = aX + b$, such particular values a^*, b^* that make $E(Y-aX+b)^2$ as small as possible. We denote $g_0^*(X) = a^*X + b^* = f_0(X)$ as the best linear approximation of Y and, clearly, since this is an approximation in a smaller class, it holds that $E(Y-f(X))^2 \leq E(Y-f_0(X))^2$. (*)

The squared correlation quotient (or correlation ratio) between X and Y is defined as $\eta^2 = 1 - \frac{E(Y-f(X))^2}{E(Y-EY)^2}$ and it happens that the squarred (usual) correlation coefficient ρ^2 satisfies $\rho^2 = 1 - \frac{E(Y-f_0(X))^2}{E(Y-EY)^2}$ (more about justifying these claims will follow later). Because of the shown relationship (*), it follows that $0 \leq \rho^2 \leq \eta^2 \leq 1$ holds.

It is clear that η^2 is a <u>"better"</u> measure of association than ρ^2 is: think about the following situation: In the scatterplot below, the best linear fit looks as the line $y = \text{const}$ no matter what X which means that $\rho^2 = 0$ but clearly a nonlinear (perhaps parabolic)



$g^*(X) = f(X)$ is much more appropriate to describe their association // and η^2 would certainly be > 0.

However, given data (X_i, Y_i), $i = 1, 2, \ldots, n$ only, the correlation coefficient

$$\hat{\rho} = \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \cdot \frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

can be evaluated easily and routinely, whereas the estimation of the correlation ratio turns out to be more challenging.